

Rolando Miranda, BS.

NYU Tandon Mechanical Engineering Graduate Student

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Portfolio: <https://rolando-miranda.github.io/>

Based in Bronx, NY

Background and Summary

Mechanical Engineer with experience in automated process development, high-volume manufacturing and coursework in mechatronics. Combines technical expertise in mechatronics, mechanical engineering, project management and lean manufacturing. Professional background in Class III medical device manufacturing, compliance with quality standards, test method design and validation. Currently functioning as a researcher supporting design and automatic control of electrochemical reactors.

Education

- **NYU Tandon School of Engineering:** MS. in Mechanical Engineering. Coursework: **Mechatronics, Robotics, Thermal Engineering.** Graduation: May 2027
- **Latin American University of Science and Technology (ULACIT):** Master's in Project Management. Graduated Suma Cum Laude (September 2025) Completed while working full-time at Terumo-Neuro
- **Universidad de Costa Rica:** Licenciature's with Honors in Mechanical Engineering. GPA 9.71 out of 10. Graduated in April 2023
- **Universidad de Costa Rica:** Bachelor's with Honors in Mechanical Engineering. GPA 9.71 out of 10. Graduated in April 2021

Work Experience

- NYU Tandon School of Engineering, **Research Assistant, Bui Lab (Electrochemistry)** (5/2026 - current)
 - Deliver power supply, mechanical and electronic subsystem design for automated reactor actuation.
 - Collaborate with chemical engineers to translate user requirements into design criteria.
 - Develop software to automatize reactor actuation and data gathering.
- NYU Tandon School of Engineering, **Graduate Course Assistant, Thermo-fluids Laboratory** (9/2025 – 5/2026)
 - Designed and built equipment for thermal system evaluation for channel flow applications from scratch for a new laboratory.
 - Assist students in their projects related to thermal system modeling, improvement and testing.
 - Draft laboratory SOPs, run laboratories and ensure correct practice execution.
- Terumo-Neuro, **Manufacturing Engineer II** (6/2023 – 7/2025):
 - Managed cross-functional validation project that doubled production capacity for highest-volume microcatheter, coordinating R&D, QA, Manufacturing, and Regulatory teams through qualification protocol development and execution.
 - Executed raw material validation strategy across five product lines partnering with R&D, reducing \$3M in backorders through systematic design verification (DV), process validation (PV) and stakeholder alignment.
 - Exceeded corporate yield targets across assigned product lines by standardizing SOPs and aligning production, quality, and management teams on process improvements.
- MicroVention-Terumo, **Manufacturing Engineer I** (9/2021 – 6/2023)
 - Performed hands-on functional testing of Class III medical devices for design & process validation.
 - Authored validation documentation collaborating across R&D, QA, and Manufacturing to achieve process qualification.
- MicroVention-Terumo, **Manufacturing Engineering Intern** (5/2021 – 9/2021)
 - Led small transfer projects due to line relocation.
 - Collaborated in Root Cause Analysis (RCA) for manufacturing issues.
 - Responsible of high-value raw material scrap and handling.
- Junior Engineer, **CIRE-TOP Group Engineering Consulting Firm** (1/2021 – 5-2021)
 - Ensured safety of high-pressure steam boilers with in-field steel thickness measurements and pressure vessel solid mechanics.
 - Executed design tasks for piping systems for energy and mining applications.

Key Skills

- **Technical Skills:**
 - Software:** MATLAB, Simulink, SolidWorks, AutoCAD, Minitab, MS Project, Power BI.
 - Manufacturing:** GxP compliance, process validation, Failure Mode and Effect Analysis (FMEA).
 - Engineering:** FEA fundamentals, thermal system modeling, mechatronics, control systems, mechanical design, robotics simulation.
 - Prototyping:** Arduino, embedded systems, 3D printing, electronic circuit design and mechatronic integration.
 - Programming:** Optimization, numerical computation and visualization with Python. Familiar with C for embedded systems.
 - AI Tools:** Use AI coding assistants (Copilot, Claude) to accelerate scripting/debugging, and LLM tools for technical writing, research synthesis, and process documentation.
- **Methodologies:**
 - Project Management:** Agile, Waterfall, Risk Management
 - Quality:** Lean Manufacturing, Six Sigma, Root Cause Analysis
- **Languages:** English Fluency (TOEFL 105), Native Spanish, German (Basic A2)